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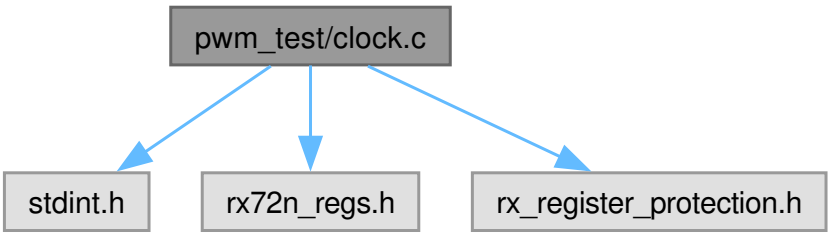
```
#include <stdint.h>
#include "rx72n_regs.h"
#include "rx_register_protection.h"
```



*

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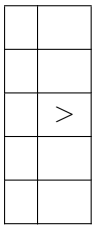
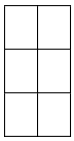
>>>

enum clock_byte_constants_t : uint8_t



```
00075             : uint8_t {
00076     k_hococr_running   = 0x00U, /**< HOCOCR: 0 = HOCO running */
00077     k_pllcr2_enable    = 0x00U, /**< PLLCR2: 0 = PLL running (inverted logic) */
00078     k_memwait_one      = 0x01U, /**< MEMWAIT: 1 wait state for ICLK > 120 MHz */
00079     k_oscovfsr_hcovf   = 0x08U, /**< OSCOVFSR bit 3: HOCO stable */
00080     k_oscovfsr_plovf   = 0x04U, /**< OSCOVFSR bit 2: PLL stable */
00081 } clock_byte_constants_t;
```

enum clock_constants_word_t : uint16_t



```
00043             : uint16_t {
00044     /*
00045      * PLLCR layout (16 bits):
00046      *   bits [1:0]  PLIDIV  : 00=/1, 01=/2, 10=/3
00047      *   bit  [4]    PLLSRCSEL: 0=MOSC, 1=HOCO
00048      *   bits [13:8] STC     : multiply = (STC + 1) / 2
00049      *
00050      * HOCO 16 MHz / 1 * 12 = 192 MHz PLL output.
00051      * STC = (12 * 2) - 1 = 23 = 0x17.
00052      * PLLSRCSEL = 1 (HOCO) -> bit 4 set.
00053      */
00054     k_pllcr_hoco_x12 = 0x1710U,
00055
00056     /*
00057      * SCKCR2: bits [7:4] UCK = /(N+1), bits [3:0] reserved (must be 0x1).
00058      * UCK = 0x3 -> /4 -> 192 MHz / 4 = 48 MHz exact.
00059      */
00060     k_sckcr2_uck_div4 = 0x0031U,
00061
00062     /* PACKCR: bit 0 UPLLSEL -- 0 routes UCLK divider input from main PLL. */
00063     k_packcr_pll_source = 0x0000U,
00064 } clock_constants_word_t;
```

enum clock_extra_addrs_t : uintptr_t

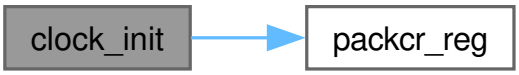
```
00036             : uintptr_t {
00037     k_packcr_addr = 0x00080044U, /**< Peripheral clock source: 0=PLL, 1=PPLL */
00038 } clock_extra_addrs_t;
```

enum clock_sckcr_t : uint32_t

```
00066             : uint32_t {
00067     /*
00068      * SCKCR dividers:
00069      *   FCK=/4 (48), ICK=/2 (96), PSTOP1=1, PSTOP0=1, BCK=/4 (48),
00070      *   PCKA=/2 (96), PCKB=/4 (48), PCKC=/2 (96), PCKD=/2 (96).
00071      */
00072     k_sckcr_value = 0x21C21211U,
00073 } clock_sckcr_t;
```

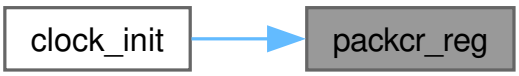
void clock_init (
void)

```
00101 {
00102     volatile rx_system_regs_t *sys = system_regs();
00103
00104     /* Step 0: unlock PRC0 (clock) + PRC1 (module stop) */
00105     *prcr_reg() = k_rx_prcr_unlock_prc0_prc1;
00106
00107     /* Step 1: start HOCO if not already running. */
00108     sys->hococr = k_hococr_running;
00109
00110     /* Step 2: wait for HOCO to stabilise. */
00111     while ((sys->oscovfsr & k_oscovfsr_hcovf) == 0U) {
00112         ___asm___ volatile("nop");
00113     }
00114
00115     /* Step 3: program PLL: HOCO 16 MHz / 1 * 12 = 192 MHz, and enable. */
00116     sys->pllcr  = k_pllcr_hoco_x12;
00117     sys->pllcr2 = k_pllcr2_enable;
00118
00119     /* Step 4: wait for PLL to stabilise. */
00120     while ((sys->oscovfsr & k_oscovfsr_plovf) == 0U) {
00121         ___asm___ volatile("nop");
00122     }
00123
00124     /* Step 5: MANDATORY -- raise flash wait state before ICLK exceeds 120 MHz. */
00125     *memwait_reg() = k_memwait_one;
00126
00127     /* Step 6: program all bus dividers. */
00128     sys->sckcr = k_sckcr_value;
00129
00130     /* Step 7: select main PLL as the UCLK divider source (UPLLSEL=0). */
00131     *packcr_reg() = k_packcr_pll_source;
00132
00133     /* Step 8: program the USB clock divider. */
00134     sys->sckcr2 = k_sckcr2_uck_div4;
00135
00136     /* Step 9: switch the system clock source to PLL. */
00137     sys->sckcr3 = k_sckcr3_cksel_pll;
00138
00139     /* Step 10: relock PRCR. */
00140     *prcr_reg() = k_rx_prcr_lock;
00141 }
```



volatile uint16_t * packcr_reg (
void) [inline], [static]

```
00087 {
00088     return (volatile uint16_t *)k_packcr_addr;
00089 }
```



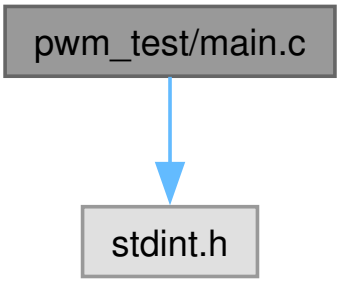
```
00001 /**
00002  * @file clock.c
00003  * @brief RX72N clock initialisation: HOCO 16 MHz -> PLL 192 -> ICLK 96 / PCKB 48.
00004  *
00005  * @details
00006  * This test sources the PLL from the internal HOCO (16 MHz) instead of
00007  * the STAR PCB's 24 MHz external crystal on MOSC/EXTAL. The choice is
00008  * deliberate (isolate the PLL bring-up from crystal startup), not because
00009  * the board lacks a crystal -- it has one.
00010  *
00011  * Target clocks:
00012  *   - PLL   = HOCO 16 MHz x 12 = 192 MHz
00013  *   - ICLK  = 192 / 2  =  96 MHz  (CPU)
00014  *   - PCKA  = 192 / 2  =  96 MHz
00015  *   - PCKB  = 192 / 4  =  48 MHz  (CMT0 ThreadX tick)
00016  *   - PCKC  = 192 / 2  =  96 MHz
00017  *   - PCKD  = 192 / 2  =  96 MHz
00018  *   - FCLK  = 192 / 4  =  48 MHz  (flash)
00019  *   - BCLK  = 192 / 4  =  48 MHz
00020  *   - UCLK  = 192 / 4  =  48 MHz  (not used, but set for completeness)
00021  *
00022  * Path: HOCO 16 MHz -> PLIDIV /1 -> STC x12 -> 192 MHz PLL output.
00023  *
00024  * SPDX-License-Identifier: MIT
00025  * @copyright Copyright (c) 2026 Locked Inc.
00026  */
00027
00028 #include <stdint.h>
00029
00030 #include "rx72n_regs.h"
00031 #include "rx_register_protection.h"
00032
00033 /*
       ============================================================================
00034  * Register addresses not exposed by the rx_system_regs_t struct
00035  *
       ============================================================================
       */
00036 typedef enum : uintptr_t {
00037     k_packcr_addr = 0x00080044U, /**< Peripheral clock source: 0=PLL, 1=PPLL */
00038 } clock_extra_addrs_t;
00039
00040 /*
       ============================================================================
00041  * Clock configuration constants
00042  *
       ============================================================================
       */
00043 typedef enum : uint16_t {
00044     /*
00045      * PLLCR layout (16 bits):
00046      *   bits [1:0]   PLIDIV  : 00=/1, 01=/2, 10=/3
00047      *   bit  [4]     PLLSRCSEL: 0=MOSC, 1=HOCO
00048      *   bits [13:8] STC      : multiply = (STC + 1) / 2
00049      *
00050      * HOCO 16 MHz / 1 * 12 = 192 MHz PLL output.
00051      * STC = (12 * 2) - 1 = 23 = 0x17.
00052      * PLLSRCSEL = 1 (HOCO) -> bit 4 set.
00053      */
00054     k_pllcr_hoco_x12 = 0x1710U,
00055
00056     /*
00057      * SCKCR2: bits [7:4] UCK = /(N+1), bits [3:0] reserved (must be 0x1).
00058      * UCK = 0x3 -> /4 -> 192 MHz / 4 = 48 MHz exact.
00059      */
00060     k_sckcr2_uck_div4 = 0x0031U,
00061
00062     /* PACKCR: bit 0 UPLLSEL -- 0 routes UCLK divider input from main PLL. */
00063     k_packcr_pll_source = 0x0000U,
00064 } clock_constants_word_t;
00065
00066 typedef enum : uint32_t {
00067     /*
00068      * SCKCR dividers:
00069      *   FCK=/4 (48), ICK=/2 (96), PSTOP1=1, PSTOP0=1, BCK=/4 (48),
00070      *   PCKA=/2 (96), PCKB=/4 (48), PCKC=/2 (96), PCKD=/2 (96).
00071      */
00072     k_sckcr_value = 0x21C21211U,
00073 } clock_sckcr_t;
00074
00075 typedef enum : uint8_t {
00076     k_hococr_running   = 0x00U, /**< HOCOCR: 0 = HOCO running */
00077     k_pllcr2_enable    = 0x00U, /**< PLLCR2: 0 = PLL running (inverted logic) */
00078     k_memwait_one      = 0x01U, /**< MEMWAIT: 1 wait state for ICLK > 120 MHz */
00079     k_oscovfsr_hcovf   = 0x08U, /**< OSCOVFSR bit 3: HOCO stable */
00080     k_oscovfsr_plovf   = 0x04U, /**< OSCOVFSR bit 2: PLL stable */
```

```
00081 } clock_byte_constants_t;
00082
00083 /*
      ============================================================================
00084  * Inline accessors
00085  *
      ============================================================================
      */
00086 static inline volatile uint16_t *packcr_reg(void)
00087 {
00088     return (volatile uint16_t *)k_packcr_addr;
00089 }
00090
00091 /*
      ============================================================================
00092  * clock_init -- bring the chip from LOCO 240 kHz to PLL 192/96/48 MHz.
00093  *
00094  * Uses HOCO (16 MHz internal oscillator) as the PLL source.
00095  * No external crystal required.
00096  *
00097  * Must run before any peripheral that depends on PCKB is touched.
00098  * Called from main() before cmt0_init() and before tx_kernel_enter().
00099  *
      ============================================================================
      */
00100 void clock_init(void)
00101 {
00102     volatile rx_system_regs_t *sys = system_regs();
00103
00104     /* Step 0: unlock PRC0 (clock) + PRC1 (module stop) */
00105     *prcr_reg() = k_rx_prcr_unlock_prc0_prc1;
00106
00107     /* Step 1: start HOCO if not already running. */
00108     sys->hococr = k_hococr_running;
00109
00110     /* Step 2: wait for HOCO to stabilise. */
00111     while ((sys->oscovfsr & k_oscovfsr_hcovf) == 0U) {
00112         ___asm___ volatile("nop");
00113     }
00114
00115     /* Step 3: program PLL: HOCO 16 MHz / 1 * 12 = 192 MHz, and enable. */
00116     sys->pllcr  = k_pllcr_hoco_x12;
00117     sys->pllcr2 = k_pllcr2_enable;
00118
00119     /* Step 4: wait for PLL to stabilise. */
00120     while ((sys->oscovfsr & k_oscovfsr_plovf) == 0U) {
00121         ___asm___ volatile("nop");
00122     }
00123
00124     /* Step 5: MANDATORY -- raise flash wait state before ICLK exceeds 120 MHz. */
00125     *memwait_reg() = k_memwait_one;
00126
00127     /* Step 6: program all bus dividers. */
00128     sys->sckcr = k_sckcr_value;
00129
00130     /* Step 7: select main PLL as the UCLK divider source (UPLLSEL=0). */
00131     *packcr_reg() = k_packcr_pll_source;
00132
00133     /* Step 8: program the USB clock divider. */
00134     sys->sckcr2 = k_sckcr2_uck_div4;
00135
00136     /* Step 9: switch the system clock source to PLL. */
00137     sys->sckcr3 = k_sckcr3_cksel_pll;
00138
00139     /* Step 10: relock PRCR. */
00140     *prcr_reg() = k_rx_prcr_lock;
00141 }
```

```
00001 /*
00002  * gpio_test/linker.ld -- Linker script for GPIO breakout board test.
00003  *
00004  * Same memory map as usb_test: 512 KB internal SRAM, 2 MB internal ROM.
00005  * Includes Renesas-syntax sections (P, B, D, C) needed by ThreadX RXv3 port.
00006  */
00007
00008 MEMORY
00009 {
00010     RAM : ORIGIN = 0x00000004, LENGTH = 0x7FFFC   /* 512 KB - 4 bytes */
00011     ROM : ORIGIN = 0xFFE00000, LENGTH = 0x200000  /* 2 MB             */
00012 }
00013
00014 SECTIONS
```

```
00015 {
00016     /* Code -- explicit ROM base so section VMA is anchored correctly. */
00017     .text 0xFFE00000 : AT(0xFFE00000)
00018     {
00019         *(.text*)
00020         *(P)
00021         *(P )
00022         *(.rodata*)
00023         *(C)
00024         *(C )
00025         *(C )
00026         etext = .;
00027     } > ROM
00028
00029     /*
00030      * Relocatable vector table -- startup.S loads INTB with this address.
00031      * Must be 4-byte aligned; only vectors 27 (SWINT) and 28 (CMT0) are
00032      * used, but allocate 256 entries (1024 bytes) for safety.
00033      */
00034     .rvectors ALIGN(4) :
00035     {
00036         __rvectors_start = .;
00037         KEEP(*(.rvectors))
00038         __rvectors_end = .;
00039     } > ROM
00040
00041     /*
00042      * Initialised data -- LMA stays in ROM, VMA in RAM.
00043      * startup.S copies _data..edata from __mdata on boot.
00044      */
00045     __mdata = .;
00046     .data : AT(__mdata)
00047     {
00048         __data = .;
00049         *(.data*)
00050         *(D)
00051         *(D )
00052         *(D )
00053         __edata = .;
00054     } > RAM
00055
00056     /* BSS -- zeroed by startup.S */
00057     .bss :
00058     {
00059         __bss = .;
00060         *(.bss*)
00061         *(COMMON)
00062         *(B)
00063         *(B )
00064         *(B )
00065         __ebss = .;
00066         . = ALIGN(4);
00067         __end = .;
00068     } > RAM AT>RAM
00069
00070     /*
00071      * Stacks -- USP grows down from top of RAM, ISP 4 KB below that.
00072      */
00073     __ustack = ORIGIN(RAM) + LENGTH(RAM);
00074     __istack = __ustack - 0x1000;
00075
00076     /*
00077      * Exception vector table at fixed hardware address 0xFFFFFF80.
00078      * 32 entries -- all unused.
00079      */
00080     .exvectors 0xFFFFFF80 : AT(0xFFFFFF80)
00081     {
00082         LONG(0xFFFFFFFF); LONG(0xFFFFFFFF); LONG(0xFFFFFFFF); LONG(0xFFFFFFFF);
00083         LONG(0xFFFFFFFF); LONG(0xFFFFFFFF); LONG(0xFFFFFFFF); LONG(0xFFFFFFFF);
00084         LONG(0xFFFFFFFF); LONG(0xFFFFFFFF); LONG(0xFFFFFFFF); LONG(0xFFFFFFFF);
00085         LONG(0xFFFFFFFF); LONG(0xFFFFFFFF); LONG(0xFFFFFFFF); LONG(0xFFFFFFFF);
00086         LONG(0xFFFFFFFF); LONG(0xFFFFFFFF); LONG(0xFFFFFFFF); LONG(0xFFFFFFFF);
00087         LONG(0xFFFFFFFF); LONG(0xFFFFFFFF); LONG(0xFFFFFFFF); LONG(0xFFFFFFFF);
00088         LONG(0xFFFFFFFF); LONG(0xFFFFFFFF); LONG(0xFFFFFFFF); LONG(0xFFFFFFFF);
00089         LONG(0xFFFFFFFF); LONG(0xFFFFFFFF); LONG(0xFFFFFFFF);
00090     } > ROM
00091
00092     /* Fixed reset vector at 0xFFFFFFFC */
00093     .fvectors 0xFFFFFFFC : AT(0xFFFFFFFC)
00094     {
00095         KEEP(*(.fvectors))
00096     } > ROM
00097 }
```

#include <stdint.h>



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```
#define GPTW0_BASE 0x000C2000U

#define GTBER (GPTW0_BASE + 0x40U) /* buffer enable */

#define GTCCRA (GPTW0_BASE + 0x4CU) /* duty A */

#define GTCCRB (GPTW0_BASE + 0x50U) /* duty B */

#define GTCNT (GPTW0_BASE + 0x48U) /* counter */

#define GTCR (GPTW0_BASE + 0x2CU) /* 32b: MD, TPCS, CST */

#define GTINTAD (GPTW0_BASE + 0x38U)

#define GTIOR (GPTW0_BASE + 0x34U) /* 32b: GTIOCA/B output config */

#define GTPBR (GPTW0_BASE + 0x68U) /* period buffer */

#define GTPR (GPTW0_BASE + 0x64U) /* period */
```

#define GTST (GPTW0_BASE + 0x3CU)

#define GTUDDTYC (GPTW0_BASE + 0x30U) /* 32b: count direction */

#define GTWP (GPTW0_BASE + 0x00U) /* 32b: write protection */

#define MSTPCRA 0x00080010U /* bit 7 = GPTW (per Renesas iodefine.h -- MSTPCRC.MSTPC6 is actually ECCRAM) */

#define P17PFS 0x0008C14FU /* pin 38 on Tom board */

#define P23PFS 0x0008C153U /* pin 34 on Tom board */

#define PA7_MASK 0x80U

#define PORT1_PDR 0x0008C001U

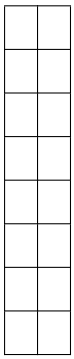
#define PORT1_PMR 0x0008C061U

#define PORT1_PODR 0x0008C021U

```
#define PORT2_PDR 0x0008C002U

#define PORT2_PMR 0x0008C062U

#define PORT2_PODR 0x0008C022U

#define PORTA_PDR 0x0008C00AU

#define PORTA_PODR 0x0008C02AU

#define PRCR 0x000803FEU

#define PWPR 0x0008C11FU

#define REG16(
                      a)

 (*(volatile uint16_t *)(a))

#define REG32(
                      a)

 (*(volatile uint32_t *)(a))
```

```
#define REG8(
                        a)


 (*(volatile uint8_t  *)(a))
```

```
enum pwm_const_t : uint32_t
```



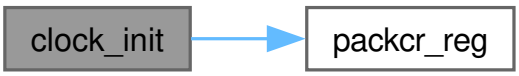
```
 00086             : uint32_t {
 00087     k_pwm_pcka_hz     = 96000000U,
 00088     k_pwm_freq_hz     = 20000U,
 00089     k_pwm_period_cnt  = 4800U,        /* 96 MHz / 20 kHz */
 00090     k_pwm_gtpr        = 4799U,        /* period - 1 */
 00091     k_duty_step       = 48U,          /* ~1% of period */
 00092     k_duty_hold_loops = 40000U,       /* delay between duty steps */
 00093     k_psel_gtiocxa    = 0x06U,        /* GTIOC0A, GTIOC0B alt function */
 00094     k_psel_gtiocxb    = 0x06U,
 00095 } pwm_const_t;
```

```
void _tx_thread_context_restore (
                        void )


 00357 {}
```

```
void _tx_thread_context_save (
                        void )


 00356 {}
```

void clock_init (
void) [extern]

```
00101 {
00102     volatile rx_system_regs_t *sys = system_regs();
00103
00104     /* Step 0: unlock PRC0 (clock) + PRC1 (module stop) */
00105     *prcr_reg() = k_rx_prcr_unlock_prc0_prc1;
00106
00107     /* Step 1: start HOCO if not already running. */
00108     sys->hococr = k_hococr_running;
00109
00110     /* Step 2: wait for HOCO to stabilise. */
00111     while ((sys->oscovfsr & k_oscovfsr_hcovf) == 0U) {
00112         ___asm___ volatile("nop");
00113     }
00114
00115     /* Step 3: program PLL: HOCO 16 MHz / 1 * 12 = 192 MHz, and enable. */
00116     sys->pllcr  = k_pllcr_hoco_x12;
00117     sys->pllcr2 = k_pllcr2_enable;
00118
00119     /* Step 4: wait for PLL to stabilise. */
00120     while ((sys->oscovfsr & k_oscovfsr_plovf) == 0U) {
00121         ___asm___ volatile("nop");
00122     }
00123
00124     /* Step 5: MANDATORY -- raise flash wait state before ICLK exceeds 120 MHz. */
00125     *memwait_reg() = k_memwait_one;
00126
00127     /* Step 6: program all bus dividers. */
00128     sys->sckcr = k_sckcr_value;
00129
00130     /* Step 7: select main PLL as the UCLK divider source (UPLLSEL=0). */
00131     *packcr_reg() = k_packcr_pll_source;
00132
00133     /* Step 8: program the USB clock divider. */
00134     sys->sckcr2 = k_sckcr2_uck_div4;
00135
00136     /* Step 9: switch the system clock source to PLL. */
00137     sys->sckcr3 = k_sckcr3_cksel_pll;
00138
00139     /* Step 10: relock PRCR. */
00140     *prcr_reg() = k_rx_prcr_lock;
00141 }
```



void cmt0_isr (
void)

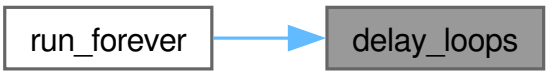
```
00358 {}
```

void dbg_pulse (
uint32_t n) [static]

```
00266 {
00267     REG8(PORTA_PDR)  |= PA7_MASK;
00268     for (uint32_t i = 0; i < n; i++) {
00269         REG8(PORTA_PODR) |= PA7_MASK;
00270         for (volatile uint32_t d = 0; d < 20000U; d++) { ___asm___ volatile("nop"); }
00271         REG8(PORTA_PODR) &= (uint8_t)~PA7_MASK;
00272         for (volatile uint32_t d = 0; d < 20000U; d++) { ___asm___ volatile("nop"); }
00273     }
00274     for (volatile uint32_t d = 0; d < 100000U; d++) { ___asm___ volatile("nop"); }
00275 }
```

void delay_loops (
uint32_t n) [inline], [static]

```
00098 {
00099     for (volatile uint32_t i = 0U; i < n; i++) {
00100         __asm__ volatile("nop");
00101     }
00102 }
```



void gptw0_init_pwm (
void) [static]

```
00139 {
00140     /* Unlock GTWP write-protection (key 0xA500 in upper bits, WP bits = 0). */
00141     REG32(GTWP)    = 0x0000A500U;
00142
00143     /* Stop counter before configuring. */
00144     REG32(GTCR)    = 0x00000000U;
00145     REG32(GTUDDTYC)= 0x00000003U;        /* UD=1 (count up) + UDF=1 (forced update) -- without UDF, UD write
        is ignored */
00146     REG32(GTCNT)   = 0x00000000U;
00147
00148     /* Period: 4800 cycles at 96 MHz = 20 kHz. GTPR = period - 1. */
00149     REG32(GTPR)    = (uint32_t)k_pwm_gtpr;
00150     REG32(GTPBR)   = (uint32_t)k_pwm_gtpr;
00151
00152     /* Start at 50% duty for initial visibility. */
00153     REG32(GTCCRA)  = (uint32_t)(k_pwm_period_cnt / 2U);
00154     REG32(GTCCRB)  = (uint32_t)(k_pwm_period_cnt - (k_pwm_period_cnt / 2U));
00155
00156     /* GTIOR per rx72n_gptw_regs.h:
00157      *   GTIOA[4:0] = 0x09  -> initial LOW, high at cycle end, low at CMA
00158      *   OAE bit 8  = 1     -> output enable on GTIOCA
00159      *   GTIOB[20:16] = 0x09 same pattern
00160      *   OBE bit 24 = 1     -> output enable on GTIOCB
00161      * (Complementary waveform is produced by setting GTCCRB = period-duty.)
00162      */
00163     /* GTIOA/GTIOB = 0x09: initial LOW, HIGH at cycle end (overflow), LOW
00164      * at compare match -> active-high sawtooth PWM, confirmed by HAL
00165      * k_gptw_gtior_oa_init_low and RX72N hw manual Ch.26 Table 26.22. */
00166     const uint32_t gtior =
00167         (0x09U << 0)  | (1U << 8)  |   /* GTIOCA */
00168         (0x09U << 16) | (1U << 24);    /* GTIOCB */
00169     REG32(GTIOR)   = gtior;
00170
00171     /* No buffering.  GTCCRA/GTCCRB writes take effect immediately. */
00172     REG32(GTBER)   = 0x00000000U;
00173
00174     /* Start counter: MD=0 saw-wave, TPCS=0 (PCLKA/1), CST=1. */
00175     REG32(GTCR)    = (0U << 16) | (0U << 24) | (1U << 0);
00176 }
```

void gptw0_pin_and_module_setup (
void) [static]

```
00108 {
00109     /* Unlock PRC1 (module stop), clear MSTPCRA.MSTPA7 (GPTW), lock. */
00110     REG16(PRCR) = 0xA502U;
00111     REG32(MSTPCRA) &= ~(1UL << 7);
00112     REG16(PRCR) = 0xA500U;
00113
00114     /* Enable PFS writes via MPC PWPR. */
00115     REG8(PWPR) = 0x00U;   /* clear B0WI */
00116     REG8(PWPR) = 0x40U;   /* set PFSWE */
00117
00118     /* P23 -> GTIOC0A, P17 -> GTIOC0B.  PSEL = 0b011110 = 0x1E per
00119      * RX72N Hardware Manual Ch.23 Table 23.4 (P17) and Table 23.6 (P23).
00120      * The HAL's 0x14 is wrong for these specific pins -- 0x14 is reserved
00121      * on P17/P23 and leaves them Hi-Z. */
00122     REG8(P23PFS) = 0x1EU;
00123     REG8(P17PFS) = 0x1EU;
00124
00125     /* Re-protect PFS. */
00126     REG8(PWPR) = 0x00U;
00127     REG8(PWPR) = 0x80U;
00128
00129     /* Switch pins to peripheral mode (PMR=1). PDR stays input side
00130      * because the GTIOC output enable in GTIOR drives the pad. */
00131     REG8(PORT2_PMR) |= (uint8_t)(1U << 3); /* P23 */
00132     REG8(PORT1_PMR) |= (uint8_t)(1U << 7); /* P17 */
00133 }
```

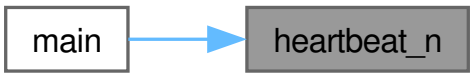
void gptw_diag_dump (
void) [static]

```
00183 {
00184     volatile uint32_t cnt_a = REG32(GTCNT);
00185     for (volatile uint32_t d = 0; d < 1000U; d++) { __asm__ volatile("nop"); }
00186     volatile uint32_t cnt_b = REG32(GTCNT);
00187     volatile uint32_t cst   = REG32(GTCR) & 1U;
00188     volatile uint32_t pmr1  = REG8(PORT1_PMR) & (1U << 7);
00189     volatile uint32_t pmr2  = REG8(PORT2_PMR) & (1U << 3);
00190
00191     /* Force P17/P23 back to GPIO mode so we can encode diagnostics on them. */
00192     REG8(PORT1_PMR) &= (uint8_t)~(1U << 7);
00193     REG8(PORT2_PMR) &= (uint8_t)~(1U << 3);
00194     REG8(PORT1_PDR) |= (1U << 7);
00195     REG8(PORT2_PDR) |= (1U << 3);
00196
00197     /* Encode diagnostics on Ch1 (P17) and Ch2 (P23):
00198      *   - 5 pulses = GTCNT advanced (counter running)
00199      *   - 3 pulses = GTCNT frozen
00200      *   - 7 pulses = CST readback = 1
00201      *   - 1 pulse  = CST readback = 0
00202      *   Sequence: cnt-state pulses, pause, cst pulses, pause, pmr pulses. */
00203     uint32_t counts_advanced = (cnt_b != cnt_a);
00204     uint32_t pulses_a = counts_advanced ? 5U : 3U;
00205     uint32_t pulses_b = cst ? 7U : 1U;
00206     uint32_t pulses_c = (pmr1 != 0U && pmr2 != 0U) ? 4U : 2U;
00207
00208     for (uint32_t round = 0; round < 3U; round++) {
00209         uint32_t n = (round == 0U) ? pulses_a : (round == 1U) ? pulses_b : pulses_c;
00210         for (uint32_t i = 0; i < n; i++) {
00211             REG8(PORT1_PODR) |= (1U << 7);
```

```
00212             REG8(PORT2_PODR) |= (1U « 3);
00213             for (volatile uint32_t d = 0; d < 30000U; d++) { ___asm___ volatile("nop"); }
00214             REG8(PORT1_PODR) &= (uint8_t)~(1U « 7);
00215             REG8(PORT2_PODR) &= (uint8_t)~(1U « 3);
00216             for (volatile uint32_t d = 0; d < 30000U; d++) { ___asm___ volatile("nop"); }
00217         }
00218         for (volatile uint32_t d = 0; d < 400000U; d++) { ___asm___ volatile("nop"); }
00219     }
00220 }
```

void heartbeat_n (
uint32_t n) [static]

```
00306 {
00307     REG8(PORTA_PDR) |= PA7_MASK;
00308     for (uint32_t i = 0; i < n; i++) {
00309         REG8(PORTA_PODR) |= PA7_MASK;
00310         for (volatile uint32_t d = 0; d < 50000U; d++) { ___asm___ volatile("nop"); }
00311         REG8(PORTA_PODR) &= (uint8_t)~PA7_MASK;
00312         for (volatile uint32_t d = 0; d < 50000U; d++) { ___asm___ volatile("nop"); }
00313     }
00314     for (volatile uint32_t d = 0; d < 200000U; d++) { ___asm___ volatile("nop"); }
00315 }
```

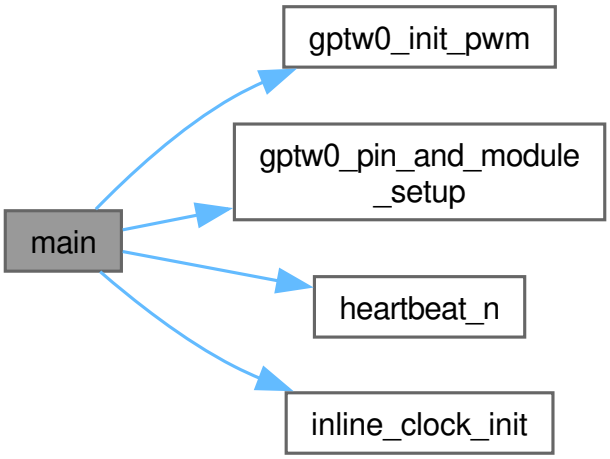


void inline_clock_init (
void) [static]

```
00283 {
00284     REG16(0x000803FEU) = 0xA503U;           /* PRC0+PRC1 unlock */
00285
00286     REG8(0x00080036U)  = 0x00U;            /* HOCOCR: start HOCO */
00287     while ((REG8(0x0008003CU) & 0x08U) == 0U) { ___asm___ volatile("nop"); } /* HCOVF */
00288
00289     REG16(0x00080028U) = 0x1710U;           /* PLLCR: HOCO*12, PLIDIV=/1 */
00290     REG8(0x0008002AU)  = 0x00U;            /* PLLCR2: enable */
00291     while ((REG8(0x0008003CU) & 0x04U) == 0U) { ___asm___ volatile("nop"); } /* PLOVF */
00292
00293     REG8(0x0008101CU)  = 0x01U;            /* MEMWAIT = 1 */
00294     REG32(0x00080020U) = 0x21C21211U;       /* SCKCR: ICLK/2 PCKA/2 PCKB/4 */
00295     REG16(0x00080044U) = 0x0001U;           /* PACKCR: bit 0 reserved must be 1 */
00296     REG16(0x00080024U) = 0x0031U;           /* SCKCR2: UCK /4 */
00297     REG16(0x00080026U) = 0x0400U;           /* SCKCR3: CKSEL=PLL */
00298
00299     REG16(0x000803FEU) = 0xA500U;           /* PRCR lock */
00300 }
```

int main (
void)

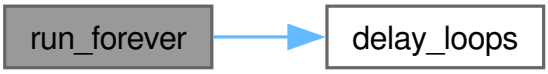
```
00318 {
00319     /* PA7 as diagnostic output; do not touch P17/P23 as GPIO after this. */
00320     REG8(PORTA_PDR)  |= PA7_MASK;
00321     REG8(PORTA_PODR) &= (uint8_t)~PA7_MASK;
00322
00323     heartbeat_n(1);                     /* 1: main() entered */
00324     inline_clock_init();
00325     heartbeat_n(2);                     /* 2: clock_init returned */
00326
00327     gptw0_pin_and_module_setup();           /* sets PMR=1 on P17+P23 */
00328     heartbeat_n(3);                     /* 3: pin/module setup done */
00329
00330     gptw0_init_pwm();                     /* configure + start counter */
00331     heartbeat_n(4);                     /* 4: GPTW init done */
00332
00333     volatile uint32_t a = REG32(GTCNT);
00334     for (volatile uint32_t d = 0; d < 10000U; d++) { __asm__ volatile("nop"); }
00335     volatile uint32_t b = REG32(GTCNT);
00336     heartbeat_n((b != a) ? 7U : 5U);          /* 7: counter running, 5: frozen */
00337
00338     /* Lock in 50% duty for clean scope capture.  No sweep. */
00339     REG32(GTCCRA) = (uint32_t)(k_pwm_period_cnt / 2U);
00340     REG32(GTCCRB) = (uint32_t)(k_pwm_period_cnt - (k_pwm_period_cnt / 2U));
00341
00342     /* PA7 heartbeat @ slow rate so we can see firmware is alive. */
00343     REG8(PORTA_PDR) |= PA7_MASK;
00344     for (;;) {
00345         REG8(PORTA_PODR) ^= PA7_MASK;
00346         for (volatile uint32_t d = 0; d < 500000U; d++) { __asm__ volatile("nop"); }
00347     }
00348     return 0;
00349 }
```



void run_forever (
void) [static]

```
00227 {
00228     /* PA7 heartbeat LED. */
00229     REG8(PORTA_PDR)  |= PA7_MASK;
00230     REG8(PORTA_PODR) &= (uint8_t)~PA7_MASK;
00231
00232     uint32_t duty      = 0U;
00233     uint32_t direction = 1U; /* 1 = up, 0 = down */
00234
00235     for (;;) {
```

```
00236         /* Update duty on both channels.  CCRB = period - CCRA keeps them
00237          * complementary (visible as mirror-image waveforms on the scope). */
00238         REG32(GTCCRA) = duty;
00239         REG32(GTCCRB) = (uint32_t)k_pwm_period_cnt - duty;
00240
00241         /* Toggle LED every loop so we can see the firmware is alive. */
00242         REG8(PORTA_PODR) ^= PA7_MASK;
00243
00244         delay_loops(k_duty_hold_loops);
00245
00246         if (direction != 0U) {
00247             duty += (uint32_t)k_duty_step;
00248             if (duty >= (uint32_t)k_pwm_period_cnt) {
00249                 duty      = (uint32_t)k_pwm_period_cnt;
00250                 direction = 0U;
00251             }
00252         } else {
00253             if (duty <= (uint32_t)k_duty_step) {
00254                 duty      = 0U;
00255                 direction = 1U;
00256             } else {
00257                 duty -= (uint32_t)k_duty_step;
00258             }
00259         }
00260     }
00261 }
```




```
00001 /**
00002  * @file main.c
00003  * @brief Minimal 20 kHz PWM output on GPTW0 for AD2 scope verification.
00004  *
00005  * @details
00006  * No motor, no H-bridge -- just drives the Motor 0 PWM pins with a
00007  * sawtooth-PWM duty sweep so the AD2 scope sees a clean 0-100% triangle
00008  * envelope on the logic-level outputs.
00009  *
00010  * Tom's PCB breakout mapping (see gpio_test/pin_map.md):
00011  *   - Tom header pin 34 -> P23 / GTIOC0A -> "IN2" (scope Ch2, blue)
00012  *   - Tom header pin 38 -> P17 / GTIOC0B -> "IN1" (scope Ch1, orange)
00013  *
00014  * Clocks (from clock.c):
00015  *   - PLL = HOCO 16 MHz * 12 = 192 MHz
00016  *   - PCKA = 192 / 2 = 96 MHz (GPTW source)
00017  *
00018  * PWM:
00019  *   - Saw-wave mode (GTCR.MD=000)
00020  *   - Period = 96 MHz / 20 kHz = 4800 counts -> GTPR = 4799
00021  *   - GTCCRA = duty (0 .. GTPR) for GTIOCA (P23)
00022  *   - GTCCRB = GTPR - duty for GTIOCB (P17) -- complementary
00023  *
00024  * Observable on the scope:
00025  *   - 50 us period (20 kHz) on both channels
00026  *   - 3.3 V logic levels (digital, not motor-driven)
00027  *   - Duty cycle sweeps 0 -> 100% -> 0% every ~4 seconds
00028  *   - Ch1 (P17) and Ch2 (P23) are mirror images
00029  *
00030  * SPDX-License-Identifier: MIT
00031  * @copyright Copyright (c) 2026 Locked Inc.
00032  */
00033
00034 #include <stdint.h>
00035
00036 extern void clock_init(void);
00037
00038 #define REG8(a)  (*(volatile uint8_t  *)(a))
00039 #define REG16(a) (*(volatile uint16_t *)(a))
00040 #define REG32(a) (*(volatile uint32_t *)(a))
00041
00042 /*
      ============================================================================
00043  * Register addresses
```

```
00044  *
       ============================================================================
       */
00045
00046 /* System / module stop */
00047 #define PRCR      0x000803FEU
00048 #define MSTPCRA   0x00080010U /* bit 7 = GPTW (per Renesas iodefine.h -- MSTPCRC.MSTPC6 is actually
       ECCRAM) */
00049
00050 /* MPC pin function select */
00051 #define PWPR      0x0008C11FU
00052 #define P17PFS    0x0008C14FU /* pin 38 on Tom board */
00053 #define P23PFS    0x0008C153U /* pin 34 on Tom board */
00054
00055 /* PORT 1 (P17) and PORT 2 (P23) peripheral mode */
00056 #define PORT1_PMR  0x0008C061U
00057 #define PORT2_PMR  0x0008C062U
00058 #define PORT1_PDR  0x0008C001U
00059 #define PORT2_PDR  0x0008C002U
00060 #define PORT1_PODR 0x0008C021U
00061 #define PORT2_PODR 0x0008C022U
00062
00063 /* GPTW0 channel registers (base 0x000C2000) */
00064 #define GPTW0_BASE 0x000C2000U
00065 #define GTWP      (GPTW0_BASE + 0x00U) /* 32b: write protection */
00066 #define GTCR      (GPTW0_BASE + 0x2CU) /* 32b: MD, TPCS, CST */
00067 #define GTUDDTYC  (GPTW0_BASE + 0x30U) /* 32b: count direction */
00068 #define GTIOR     (GPTW0_BASE + 0x34U) /* 32b: GTIOCA/B output config */
00069 #define GTINTAD   (GPTW0_BASE + 0x38U)
00070 #define GTST      (GPTW0_BASE + 0x3CU)
00071 #define GTBER     (GPTW0_BASE + 0x40U) /* buffer enable */
00072 #define GTCNT     (GPTW0_BASE + 0x48U) /* counter */
00073 #define GTCCRA    (GPTW0_BASE + 0x4CU) /* duty A */
00074 #define GTCCRB    (GPTW0_BASE + 0x50U) /* duty B */
00075 #define GTPR      (GPTW0_BASE + 0x64U) /* period */
00076 #define GTPBR     (GPTW0_BASE + 0x68U) /* period buffer */
00077
00078 /* Heartbeat LED on PA7 (pin 88) so we see firmware is alive. */
00079 #define PORTA_PDR  0x0008C00AU
00080 #define PORTA_PODR 0x0008C02AU
00081 #define PA7_MASK   0x80U
00082
00083 /*
       ============================================================================
00084  * PWM configuration constants
00085  *
       ============================================================================
       */
00086 typedef enum : uint32_t {
00087     k_pwm_pcka_hz     = 96000000U,
00088     k_pwm_freq_hz     = 20000U,
00089     k_pwm_period_cnt  = 4800U,        /* 96 MHz / 20 kHz */
00090     k_pwm_gtpr        = 4799U,        /* period - 1 */
00091     k_duty_step       = 48U,          /* ~1% of period */
00092     k_duty_hold_loops = 40000U,       /* delay between duty steps */
00093     k_psel_gtiocxa    = 0x06U,        /* GTIOC0A, GTIOC0B alt function */
00094     k_psel_gtiocxb    = 0x06U,
00095 } pwm_const_t;
00096
00097 static inline void delay_loops(uint32_t n)
00098 {
00099     for (volatile uint32_t i = 0U; i < n; i++) {
00100         __asm__ volatile("nop");
00101     }
00102 }
00103
00104 /*
       ============================================================================
00105  * Bring GPTW0 out of module stop and wire P23/P17 to GTIOC0A/GTIOC0B.
00106  *
       ============================================================================
       */
00107 static void gptw0_pin_and_module_setup(void)
00108 {
00109     /* Unlock PRC1 (module stop), clear MSTPCRA.MSTPA7 (GPTW), lock. */
00110     REG16(PRCR) = 0xA502U;
00111     REG32(MSTPCRA) &= ~(1UL << 7);
00112     REG16(PRCR) = 0xA500U;
00113
00114     /* Enable PFS writes via MPC PWPR. */
00115     REG8(PWPR) = 0x00U;   /* clear B0WI */
00116     REG8(PWPR) = 0x40U;   /* set PFSWE */
00117
00118     /* P23 -> GTIOC0A, P17 -> GTIOC0B.  PSEL = 0b011110 = 0x1E per
00119      * RX72N Hardware Manual Ch.23 Table 23.4 (P17) and Table 23.6 (P23).
00120      * The HAL's 0x14 is wrong for these specific pins -- 0x14 is reserved
00121      * on P17/P23 and leaves them Hi-Z. */
```

```
00122     REG8(P23PFS) = 0x1EU;
00123     REG8(P17PFS) = 0x1EU;
00124
00125     /* Re-protect PFS. */
00126     REG8(PWPR) = 0x00U;
00127     REG8(PWPR) = 0x80U;
00128
00129     /* Switch pins to peripheral mode (PMR=1). PDR stays input side
00130      * because the GTIOC output enable in GTIOR drives the pad. */
00131     REG8(PORT2_PMR) |= (uint8_t)(1U << 3); /* P23 */
00132     REG8(PORT1_PMR) |= (uint8_t)(1U << 7); /* P17 */
00133 }
00134
00135 /*
    ================================================================================
00136  * Configure GPTW0 as 20 kHz saw-wave PWM with complementary A/B outputs.
00137  *
    ================================================================================
    */
00138 static void gptw0_init_pwm(void)
00139 {
00140     /* Unlock GTWP write-protection (key 0xA500 in upper bits, WP bits = 0). */
00141     REG32(GTWP)    = 0x0000A500U;
00142
00143     /* Stop counter before configuring. */
00144     REG32(GTCR)    = 0x00000000U;
00145     REG32(GTUDDTYC)= 0x00000003U;        /* UD=1 (count up) + UDF=1 (forced update) -- without UDF, UD write
    is ignored */
00146     REG32(GTCNT)   = 0x00000000U;
00147
00148     /* Period: 4800 cycles at 96 MHz = 20 kHz. GTPR = period - 1. */
00149     REG32(GTPR)    = (uint32_t)k_pwm_gtpr;
00150     REG32(GTPBR)   = (uint32_t)k_pwm_gtpr;
00151
00152     /* Start at 50% duty for initial visibility. */
00153     REG32(GTCCRA)  = (uint32_t)(k_pwm_period_cnt / 2U);
00154     REG32(GTCCRB)  = (uint32_t)(k_pwm_period_cnt - (k_pwm_period_cnt / 2U));
00155
00156     /* GTIOR per rx72n_gptw_regs.h:
00157      *   GTIOA[4:0] = 0x09  -> initial LOW, high at cycle end, low at CMA
00158      *   OAE bit 8  = 1     -> output enable on GTIOCA
00159      *   GTIOB[20:16] = 0x09 same pattern
00160      *   OBE bit 24 = 1     -> output enable on GTIOCB
00161      * (Complementary waveform is produced by setting GTCCRB = period-duty.)
00162      */
00163     /* GTIOA/GTIOB = 0x09: initial LOW, HIGH at cycle end (overflow), LOW
00164      * at compare match -> active-high sawtooth PWM, confirmed by HAL
00165      * k_gptw_gtior_oa_init_low and RX72N hw manual Ch.26 Table 26.22. */
00166     const uint32_t gtior =
00167         (0x09U << 0)  | (1U << 8)  |   /* GTIOCA */
00168         (0x09U << 16) | (1U << 24);    /* GTIOCB */
00169     REG32(GTIOR)   = gtior;
00170
00171     /* No buffering.  GTCCRA/GTCCRB writes take effect immediately. */
00172     REG32(GTBER)   = 0x00000000U;
00173
00174     /* Start counter: MD=0 saw-wave, TPCS=0 (PCLKA/1), CST=1. */
00175     REG32(GTCR)    = (0U << 16) | (0U << 24) | (1U << 0);
00176 }
00177
00178 /* Debug helper: bring out two GPIOs (P17 and P23 as GPIO override) that
00179  * mirror whether GTCNT is nonzero and whether GTCR CST bit reads back.
00180  * WARNING: this re-purposes the PWM pins so scope will see GPIO-style
00181  * toggling not PWM.  Use only for debug diagnosis. */
00182 static void gptw_diag_dump(void)
00183 {
00184     volatile uint32_t cnt_a = REG32(GTCNT);
00185     for (volatile uint32_t d = 0; d < 1000U; d++) { ___asm___ volatile("nop"); }
00186     volatile uint32_t cnt_b = REG32(GTCNT);
00187     volatile uint32_t cst   = REG32(GTCR) & 1U;
00188     volatile uint32_t pmr1  = REG8(PORT1_PMR) & (1U << 7);
00189     volatile uint32_t pmr2  = REG8(PORT2_PMR) & (1U << 3);
00190
00191     /* Force P17/P23 back to GPIO mode so we can encode diagnostics on them. */
00192     REG8(PORT1_PMR) &= (uint8_t)~(1U << 7);
00193     REG8(PORT2_PMR) &= (uint8_t)~(1U << 3);
00194     REG8(PORT1_PDR) |= (1U << 7);
00195     REG8(PORT2_PDR) |= (1U << 3);
00196
00197     /* Encode diagnostics on Ch1 (P17) and Ch2 (P23):
00198      *   - 5 pulses = GTCNT advanced (counter running)
00199      *   - 3 pulses = GTCNT frozen
00200      *   - 7 pulses = CST readback = 1
00201      *   - 1 pulse  = CST readback = 0
00202      *   Sequence: cnt-state pulses, pause, cst pulses, pause, pmr pulses. */
00203     uint32_t counts_advanced = (cnt_b != cnt_a);
00204     uint32_t pulses_a = counts_advanced ? 5U : 3U;
```

```
00205     uint32_t pulses_b = cst ? 7U : 1U;
00206     uint32_t pulses_c = (pmr1 != 0U && pmr2 != 0U) ? 4U : 2U;
00207
00208     for (uint32_t round = 0; round < 3U; round++) {
00209         uint32_t n = (round == 0U) ? pulses_a : (round == 1U) ? pulses_b : pulses_c;
00210         for (uint32_t i = 0; i < n; i++) {
00211             REG8(PORT1_PODR) |= (1U << 7);
00212             REG8(PORT2_PODR) |= (1U << 3);
00213             for (volatile uint32_t d = 0; d < 30000U; d++) { __asm__ volatile("nop"); }
00214             REG8(PORT1_PODR) &= (uint8_t)~(1U << 7);
00215             REG8(PORT2_PODR) &= (uint8_t)~(1U << 3);
00216             for (volatile uint32_t d = 0; d < 30000U; d++) { __asm__ volatile("nop"); }
00217         }
00218         for (volatile uint32_t d = 0; d < 400000U; d++) { __asm__ volatile("nop"); }
00219     }
00220 }
00221
00222 /*
      ================================================================================
00223  * LED + PWM duty sweep forever.  Produces a visible 0..100% triangle on the
00224  * scope so you can watch the duty cycle move.
00225  *
      ================================================================================
      */
00226 static void run_forever(void)
00227 {
00228     /* PA7 heartbeat LED. */
00229     REG8(PORTA_PDR)  |= PA7_MASK;
00230     REG8(PORTA_PODR) &= (uint8_t)~PA7_MASK;
00231
00232     uint32_t duty      = 0U;
00233     uint32_t direction = 1U; /* 1 = up, 0 = down */
00234
00235     for (;;) {
00236         /* Update duty on both channels.  CCRB = period - CCRA keeps them
00237          * complementary (visible as mirror-image waveforms on the scope). */
00238         REG32(GTCCRA) = duty;
00239         REG32(GTCCRB) = (uint32_t)k_pwm_period_cnt - duty;
00240
00241         /* Toggle LED every loop so we can see the firmware is alive. */
00242         REG8(PORTA_PODR) ^= PA7_MASK;
00243
00244         delay_loops(k_duty_hold_loops);
00245
00246         if (direction != 0U) {
00247             duty += (uint32_t)k_duty_step;
00248             if (duty >= (uint32_t)k_pwm_period_cnt) {
00249                 duty      = (uint32_t)k_pwm_period_cnt;
00250                 direction = 0U;
00251             }
00252         } else {
00253             if (duty <= (uint32_t)k_duty_step) {
00254                 duty      = 0U;
00255                 direction = 1U;
00256             } else {
00257                 duty -= (uint32_t)k_duty_step;
00258             }
00259         }
00260     }
00261 }
00262
00263 /* Slow toggle on PA7 LED so AD2 can see how far we got.
00264  * Probe a 3rd channel to PA7 / pin 88 and count pulses. */
00265 static void dbg_pulse(uint32_t n)
00266 {
00267     REG8(PORTA_PDR)  |= PA7_MASK;
00268     for (uint32_t i = 0; i < n; i++) {
00269         REG8(PORTA_PODR) |= PA7_MASK;
00270         for (volatile uint32_t d = 0; d < 20000U; d++) { __asm__ volatile("nop"); }
00271         REG8(PORTA_PODR) &= (uint8_t)~PA7_MASK;
00272         for (volatile uint32_t d = 0; d < 20000U; d++) { __asm__ volatile("nop"); }
00273     }
00274     for (volatile uint32_t d = 0; d < 100000U; d++) { __asm__ volatile("nop"); }
00275 }
00276
00277 /*
      ================================================================================
00278  * Clock setup verified working on Tom's PCB (no external crystal).
00279  * HOCO 16 MHz internal -> PLL x12 = 192 MHz, PCKA /2 = 96 MHz.
00280  * Copied from an earlier HOCO PID bring-up test that enumerated 1209:0002 on this
00281  * hardware, so these exact register writes are known-good. */
00282 static void inline_clock_init(void)
00283 {
00284     REG16(0x000803FEU) = 0xA503U;            /* PRC0+PRC1 unlock */
00285
00286     REG8(0x00080036U)  = 0x00U;              /* HOCOCR: start HOCO */
00287     while ((REG8(0x0008003CU) & 0x08U) == 0U) { __asm__ volatile("nop"); } /* HCOVF */
```

```
00288
00289     REG16(0x00080028U) = 0x1710U;           /* PLLCR: HOCO*12, PLIDIV=/1 */
00290     REG8(0x0008002AU)  = 0x00U;             /* PLLCR2: enable */
00291     while ((REG8(0x0008003CU) & 0x04U) == 0U) { __asm__ volatile("nop"); } /* PLOVF */
00292
00293     REG8(0x0008101CU)  = 0x01U;             /* MEMWAIT = 1 */
00294     REG32(0x00080020U) = 0x21C21211U;       /* SCKCR: ICLK/2 PCKA/2 PCKB/4 */
00295     REG16(0x00080044U) = 0x0001U;           /* PACKCR: bit 0 reserved must be 1 */
00296     REG16(0x00080024U) = 0x0031U;           /* SCKCR2: UCK /4 */
00297     REG16(0x00080026U) = 0x0400U;           /* SCKCR3: CKSEL=PLL */
00298
00299     REG16(0x000803FEU) = 0xA500U;           /* PRCR lock */
00300 }
00301
00302 /* Heartbeat/diagnostic pulses on PA7 only -- do NOT touch P17/P23 after
00303  * GPTW takes them in peripheral mode, or we clobber PMR and the PWM
00304  * output stops reaching the pad. */
00305 static void heartbeat_n(uint32_t n)
00306 {
00307     REG8(PORTA_PDR) |= PA7_MASK;
00308     for (uint32_t i = 0; i < n; i++) {
00309         REG8(PORTA_PODR) |= PA7_MASK;
00310         for (volatile uint32_t d = 0; d < 50000U; d++) { __asm__ volatile("nop"); }
00311         REG8(PORTA_PODR) &= (uint8_t)~PA7_MASK;
00312         for (volatile uint32_t d = 0; d < 50000U; d++) { __asm__ volatile("nop"); }
00313     }
00314     for (volatile uint32_t d = 0; d < 200000U; d++) { __asm__ volatile("nop"); }
00315 }
00316
00317 int main(void)
00318 {
00319     /* PA7 as diagnostic output; do not touch P17/P23 as GPIO after this. */
00320     REG8(PORTA_PDR)  |= PA7_MASK;
00321     REG8(PORTA_PODR) &= (uint8_t)~PA7_MASK;
00322
00323     heartbeat_n(1);                         /* 1: main() entered */
00324     inline_clock_init();
00325     heartbeat_n(2);                         /* 2: clock_init returned */
00326
00327     gptw0_pin_and_module_setup();           /* sets PMR=1 on P17+P23 */
00328     heartbeat_n(3);                         /* 3: pin/module setup done */
00329
00330     gptw0_init_pwm();                       /* configure + start counter */
00331     heartbeat_n(4);                         /* 4: GPTW init done */
00332
00333     volatile uint32_t a = REG32(GTCNT);
00334     for (volatile uint32_t d = 0; d < 10000U; d++) { __asm__ volatile("nop"); }
00335     volatile uint32_t b = REG32(GTCNT);
00336     heartbeat_n((b != a) ? 7U : 5U);        /* 7: counter running, 5: frozen */
00337
00338     /* Lock in 50% duty for clean scope capture.  No sweep. */
00339     REG32(GTCCRA) = (uint32_t)(k_pwm_period_cnt / 2U);
00340     REG32(GTCCRB) = (uint32_t)(k_pwm_period_cnt - (k_pwm_period_cnt / 2U));
00341
00342     /* PA7 heartbeat @ slow rate so we can see firmware is alive. */
00343     REG8(PORTA_PDR) |= PA7_MASK;
00344     for (;;) {
00345         REG8(PORTA_PODR) ^= PA7_MASK;
00346         for (volatile uint32_t d = 0; d < 500000U; d++) { __asm__ volatile("nop"); }
00347     }
00348     return 0;
00349 }
00350
00351 /*
    ============================================================================
00352  * Stubs for symbols referenced by vectors.S (ThreadX ISRs, CMT0) that this
00353  * minimal firmware doesn't use. Prevents linker errors without dragging in
00354  * the full ThreadX library or a real CMT0 driver.
00355  *
    ============================================================================
    */
00356 void __tx__thread_context_save(void)    {}
00357 void __tx__thread_context_restore(void) {}
00358 void cmt0_isr(void)                     {}
```